



Machine Learning Points of Failure

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Introduction

- Evasion Attacks
- Poisoning Attacks
- Model Inversion
- Membership Inference
- Model Stealing
- Reprogramming ML
- Backdoors
- Attack ML supply chain

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Evasion attacks

- Any attack that attempts to *evade* a detection
- Most commonly done with adversarial perturbations, which are added to the input
 - targeted attack (force the model to declare a specific class)
 - untargeted attack (force the model to make any error)

Poisoning Attacks

- Adversary will *poison* the dataset, which will affect all downstream tasks like training the model
- This attack appears less likely (or realistic) – how will an adversary access our data?
 - Penetration of internal systems through classic methods
 - poison *public* datasets that are widely used
 - there were allegations that Kaspersky uploaded false positives to virus-total to confuse competitors
 - many datasets are large and openly shared – it is theoretically possible to poison the well and affect all downstream users

Model Inversion

- Using the confidence scores from models, attackers can retrieve the actual training data
- Researchers were able to accurately reconstruct faces from a model's output alone.

Model Inversion Attacks that Exploit Confidence Information and Basic Countermeasures



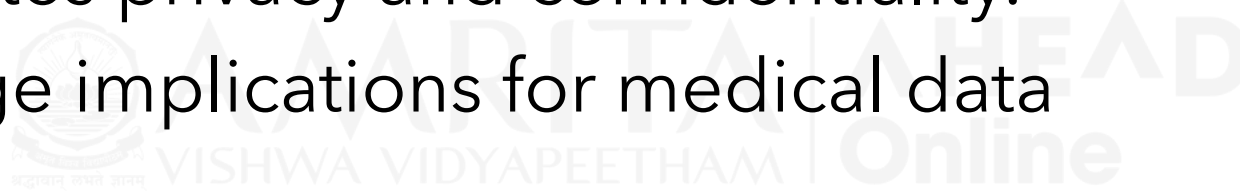
Figure 1: An image recovered using a new model inversion attack (left) and a training set image of the victim (right). The attacker is given only the person's name and access to a facial recognition system that returns a class confidence score.

<https://dl.acm.org/doi/10.1145/2810103.2813677>

Membership Inference



- Attackers are able to figure out if a data sample was part of a model's training dataset or not.
- This violates privacy and confidentiality.
- It has large implications for medical data



Model Stealing

- Similar to inversion and membership inference, model stealing is when an attacker steals the actual *model*
- This could be the weights of a neural network.
- The attackers can functionality recreate the model only by querying it.
- This affects companies ability to hold an edge in competition and rights to their intellectual property.

Reprogramming and Backdoors

- Reprogramming:
 - Attackers can repurpose a machine learning model to do something else.
 - Researchers demonstrated it by using adversarial input that repurposed an image classification model to count squares
- Backdoors
 - The training process can be corrupted to include a backdoor that is activated by a special input.
 - Example: researchers create a street sign classifier that will work like normal. Except when a special sticker is placed on stop-signs, then it will classify the stop-sign as a speed limit.

Supply Chain Attacks



- Machine learning inherently has a pipeline
- Data → preprocessing → feature extraction → model training → inference.
- Further, large models are *reused* – they are shared online (huggingface).
- An adversary could easily host a malicious model in an online repository
- The malicious model could contain a backdoor or other malicious code.

Conclusion

- Evasion
- Inversion
- Model stealing
- Reprogramming models
- Backdoors
- Supply chain attacks

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Defenses Against Evasion Attacks.

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Introduction

- Detection vs. Robustness
- Robustness
- Detection
- Defense is hard!

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Detection vs. Robustness



- In defending against evasion attacks, there are two main strategies: detection and robustness
- Detection is what it sounds like: *detect* if an input is adversarial or not.
- After detecting the malicious input, we can quarantine it or examine it further
- Robustness is the practice of making a model unaffected (or less affected) by the adversarial inputs
- In the case of robustness, the model will simply correctly classify the adversarial example



Robustness techniques

- Adversarial training
 - Create adversarial samples and give them the correct label.
 - Insert them into the training data and train!
 - This works, kinda
- Feature denoising:
 - Remove features that are strongly correlated with adversarial perturbations
 - The intuition is that certain features lead to adversarial examples – remove them, and it will be better

Detection – Feature Squeezing

- Feature squeezing
 - Make a super “sample” that is a combination of all the original images – it’s a baseline feature space
 - Compare all incoming samples to this space – if it’s very different, then call it adversarial

Detection - Uncertainty



- Measure the certainty of the model's prediction on a sample
- If the uncertainty is beyond a certain threshold, consider the sample to be adversarial
- Experiments on tabular data have shown that uncertainty is correlated to adversarial samples

Defending is Hard

- Breaking things is easy!
- Defending things is very hard
- *Many* defenses have been proposed and then later broken. Entire papers have been retracted.
- "On Evaluating Adversarial Robustness"

Conclusion

- Robustness
- Detection
- Defending is hard



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Industry Readiness

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Introduction

- Paper review:
"Adversarial Machine Learning – Industry Perspectives"
- Industry is not ready

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Conclusion

- People have little to no understanding of the risks inherent with ML
- Industry has not prepared anyone yet

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Threat Models and Secure Development in Machine Learning

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Introduction

- Microsoft SDL
- Mitre



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Conclusion

- There exist some beginnings of a framework to develop ML securely



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Penetration Testing AI Models

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Introduction

- Penetration Testing AI
- Microsoft Counterfit
- NIST- Dioptra (not ready yet)



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AHEAD
Online

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Penetration Testing AI

- Ideally we could run a suite of attacks against our AI models
- If you know the basic thing that your model is supposed to do, there are enough algorithms already written in code libraries that *you could* automate attacks against it
- That is exactly what Microsoft built us!

Conclusion

- Reviewed Counterfit
- Looked at Dioptra



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